

Bansilal Ramnath Agarwal Charitable Trust's
Vishwakarma Institute of Technology, Pune-37

(An autonomous Institute of Savitribai Phule Pune University)



Department of Multidisciplinary
Engineering

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ER Diagram & Its Components:-

An Entity - Relationship (ER) diagram is a visual representation of database's logical structure. It illustrates how data is stored, the entities (objects) that exist in system, & how these entities relate to each other. ER diagrams are one of the most widely used tools in db design because they helped developers, database administrators & stakeholders understand the system's data model before implementation.

This concept was first introduced by Peter Chen in 1976 as part of Entity - Relationship Model. These diagrams act as a blueprint for designing relational database, ensuring that all necessary data elements & their relationships are clearly defined before any coding or physical implementation takes place.

Main components of ER diagram:-

- ① Entities:-
Represent real-world objects (eg:- user, Product, etc.).
 - Strong Entity :- Exists independently.
 - Weak Entity :- Depends on strong entity.

- ② Attributes:-
Properties of entities or relationships.
 - Simple, composite, Derived or Multivalued.

③ Relationships :-

Connections between entities (shown by diamonds).

- Unary, Binary, Ternary relationships.

④ Cardinality :-

Defines how many instances of one entity relate to another.

- One-to-One ($1:1$), one-to-Many ($1:N$), Many-to-Many ($M:N$).

⑤ Participation :-

- Total : all instances participate.
- Partial : some instances participate.

⑥ Keys :-

Uniquely identify records.

- Primary & foreign key.

Advantages :-

- Clear & easy to understand.
- Detects logical errors early.
- Serves as documentation for the database.

Title:-

Perform requirement analysis in detail for the same. Draw an entity-relationship diagram for proposed database.

Description: Our project, Workout Planner, helps users track workouts and receive diet recommendations. The ER diagram shows the main entities, their attributes, and relationships. User Profile stores personal details and links to training records and diet recommendations. Equipment is linked to exercises, which connect to training records. Recipes with nutritional details link to diet recommendations for users. Standard ER symbols are used to represent the structure and connections clearly.

ER Diagram:

